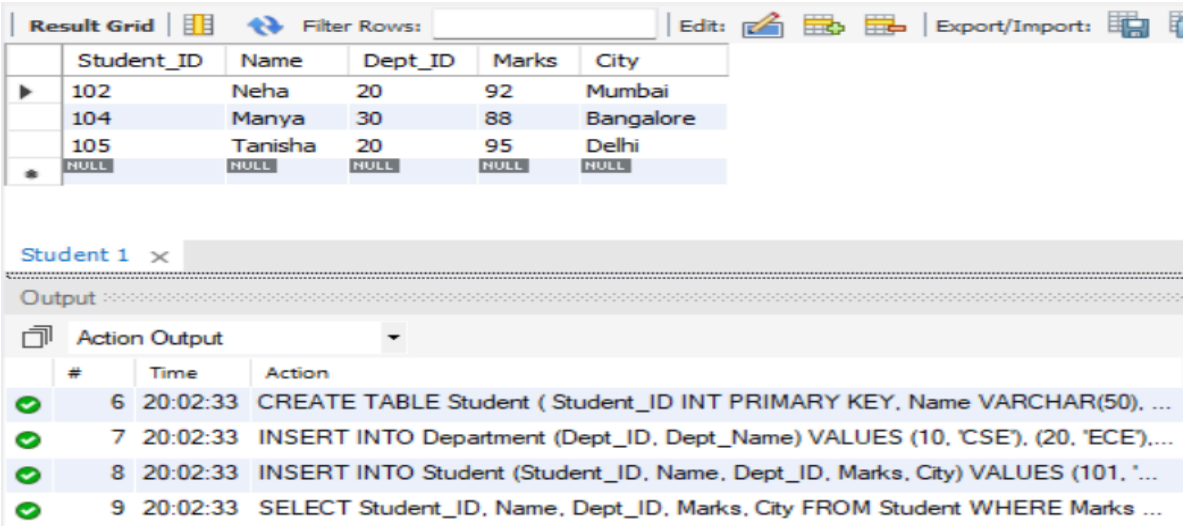
Subquery 1 — Students scoring above the Average

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (SELECT AVG(Marks) FROM Student);
```

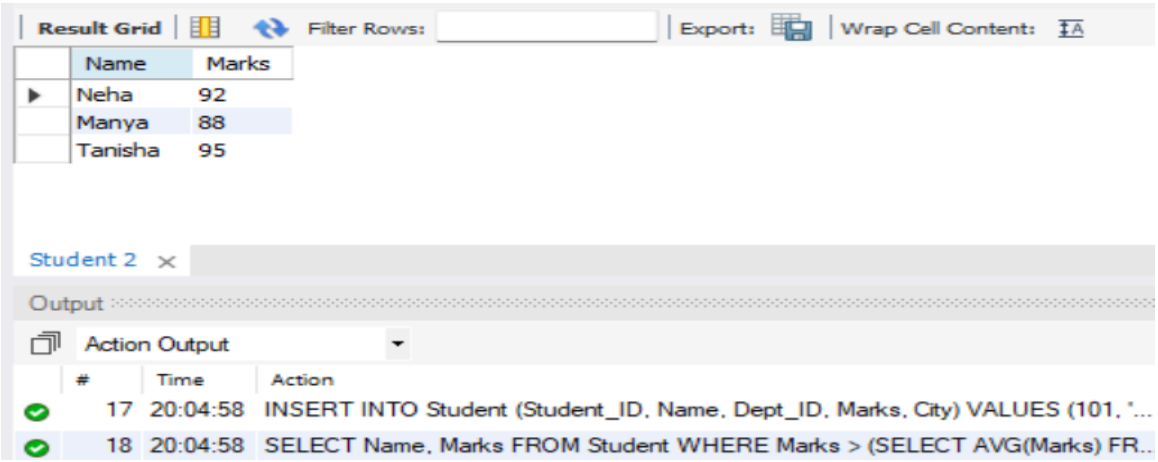


Outer:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
```

(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);

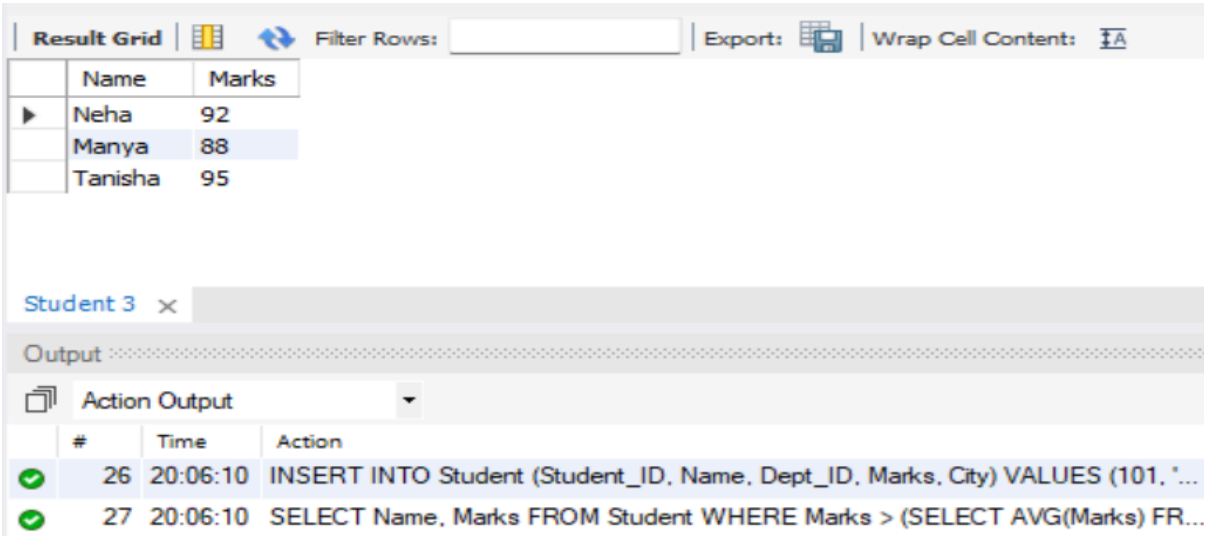


Inner:

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

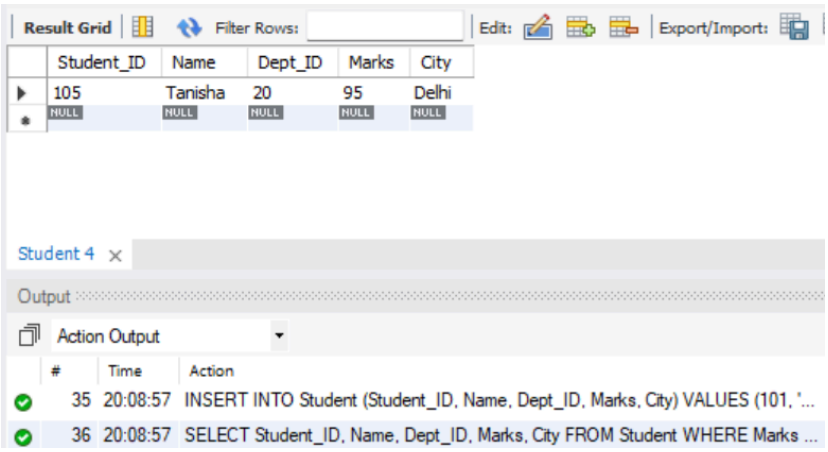
SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```



Subquery 2 — Student(s) who scored the highest marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks = (SELECT MAX(Marks) FROM Student);
```

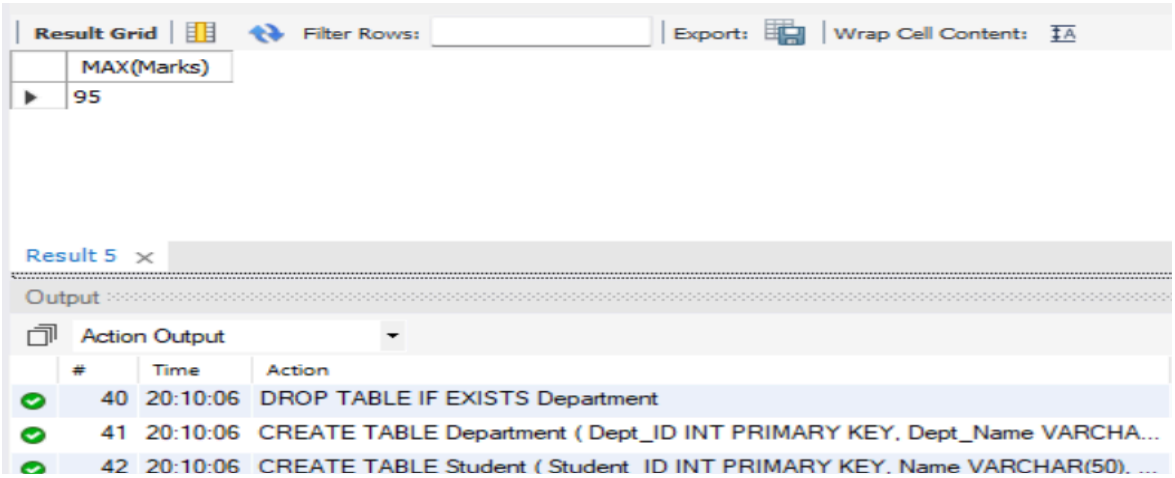


Inner:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
```

(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT MAX(Marks) FROM Student

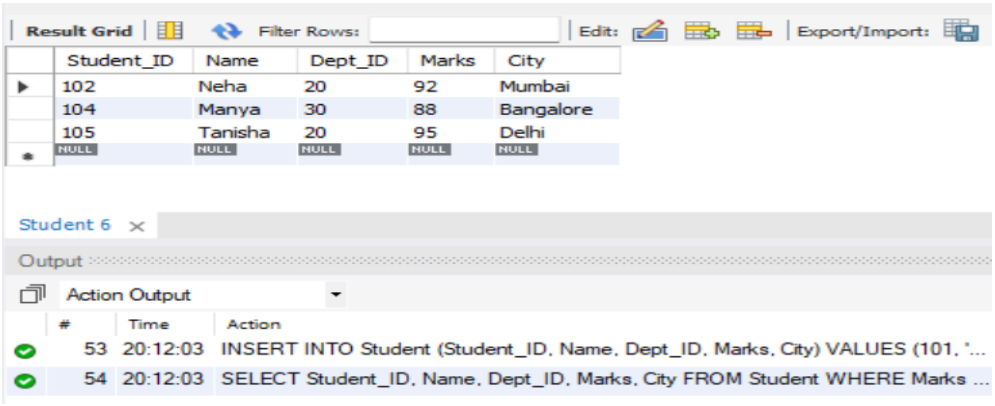


Subquery 3 — Students scoring above Rahul

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES

(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT Marks FROM Student WHERE Name = 'Rahul' OR Name = 'Navya');

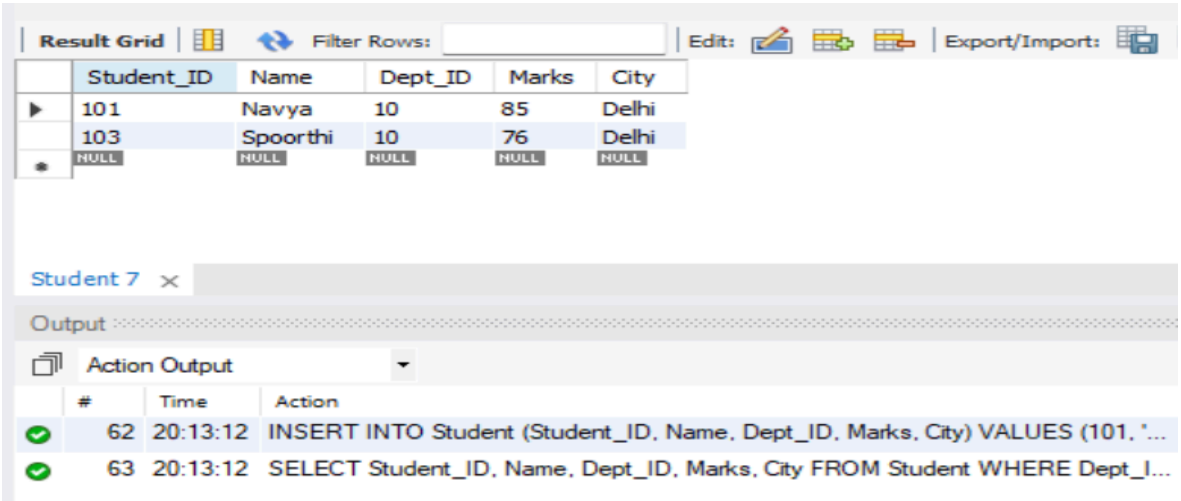


Subquery 4 — Students belonging to the same department as Rahul

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)

```
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

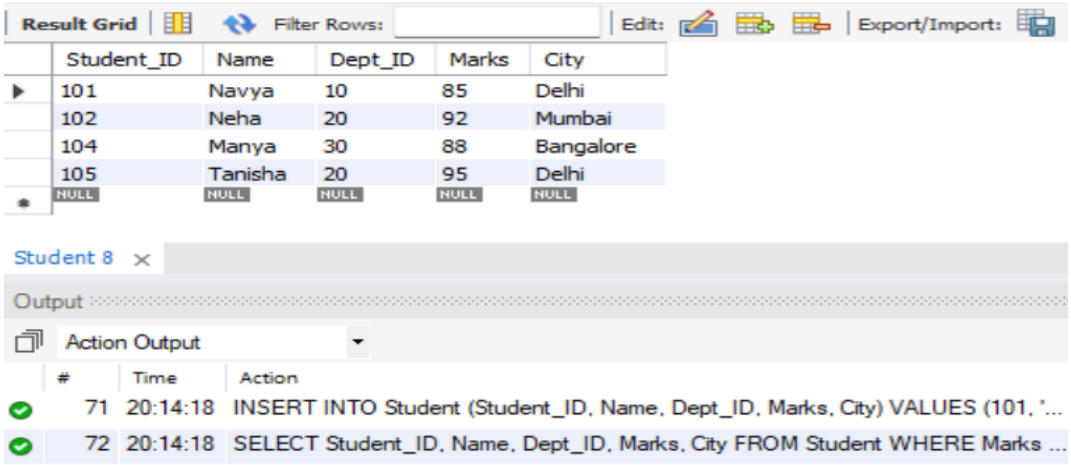
SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID =
(SELECT Dept_ID FROM Student WHERE Name = 'Navya');
```



Subquery 5 — Students who scored above the average CSE marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student WHERE Dept_ID = 10);
```

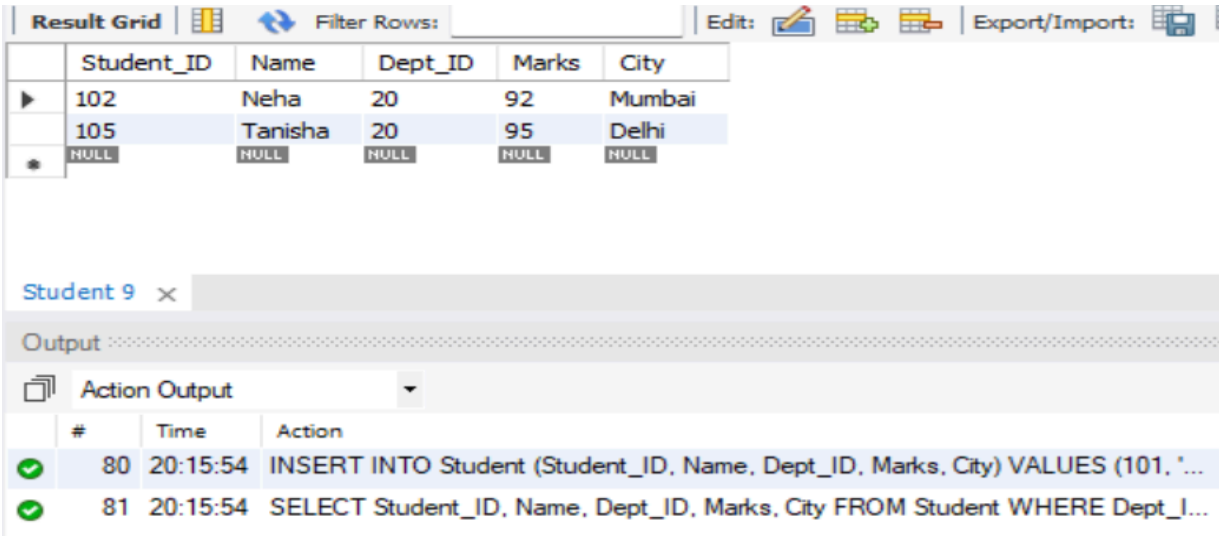


Subquery 6 — Students who belong to a department having a student from Mumbai

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
```

(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID IN
(SELECT Dept_ID FROM Student WHERE City = 'Mumbai');



Subquery 7 — Students who scored higher than every ECE student

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),

```
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (
SELECT MAX(Marks)
FROM Student
WHERE Dept_ID = 20
);
```

